

Metro systems and operations

Signalling, traction power, depots, service spacing, first and last trains, and platform doors across Taipei-region metro lines.

METRO SYSTEMS AND OPERATIONS			
TRACTION STANDARD 750 V DC third rail on Taipei [/rail/this/stations/taipei/] high- and medium-capacity metro systems ^[8]	HIGH-CAPACITY SIGNALLING Alstom/GRS conventional fixed block with audio-frequency track circuits ^[1]	WENHU SIGNALLING Bombardier CITYFLO 650 moving-block CBTC ^[2]	CIRCULAR SIGNALLING Hitachi CBTC, unattended operation ^[3]
CURRENT DOOR REGISTER TBC at station level	COMPARABLE DESIGN VERSUS ACTUAL CAPACITY TBC		

One network, several systems

The Taipei-region metro is a family of systems rather than one repeated engineering design. The high-capacity R, G, O and BL lines share an Alstom (formerly GRS) conventional fixed-block signalling family with audio-frequency track circuits. ^[1] BR uses Bombardier CITYFLO 650 moving-block CBTC, with a moving-block coordinate system and position updates at roughly half-second intervals. ^[2] Y uses Hitachi CBTC for unattended operation, and DORTS says the same signalling approach is used on LB. ^[3]

That distinction matters. Fixed block describes how train separation is protected; it does not by itself say that a line is or is not automated. The published sources therefore keep signalling block, train-control functions and operating mode as separate fields. For V and K, NTMC [\[/rail/operators/ntmc/\]](#) publishes a staffed, low-floor light-rail system with light-rail and level-crossing signalling, but does not name a signalling generation or supplier. ^[5] For A, the government project report specifies CBTC with ATP, ATS and ATO, while the train manufacturer publishes CBTC/ATP for the delivered fleet. ^[6] ^[7]

LINE	SIGNALLING EVIDENCE	POWER EVIDENCE
BR	CITYFLO 650 moving-block CBTC, Bombardier. ^[2]	750 V DC conductor rail; Wenhua's supply includes 161 kV dual circuits and 22 kV distribution. ^[10]

LINE	SIGNALLING EVIDENCE	POWER EVIDENCE
R, G, O, BL	Alstom/GRS conventional fixed block and audio-frequency track circuits. ^[1]	Taipei Metro [/rail/operators/trtc/] 750 V DC third rail. ^[8]
Y	Hitachi CBTC, driverless operation. ^[3]	750 V DC third rail. ^[5]
LB	CBTC family and driverless train design are established; supplier and generation are TBC. ^[3] ^[5]	750 V DC third rail. ^[5]
V, K	Staffed light rail with level-crossing signalling; supplier and generation are TBC. ^[5]	750 V DC overhead/energy-storage system. ^[5]
A	CBTC with ATP/ATS/ATO is documented; a single current generation label is TBC. ^[6] ^[7]	750 V DC third rail. ^[7]

Traction power and substations

DORTS describes 750 V DC third rail as the Taipei Metro traction standard. ^[8] The normal conversion chain is 22 kV AC through transformer and rectifier equipment; the manual distinguishes 24-pulse equipment on high-capacity systems from the 12-pulse equipment used generally on medium-capacity systems. ^[8] These are system design rules, not a current station-by-station equipment inventory. DORTS gives approximate traction-substation spacing of 1.7–2 km for high-capacity lines and 1–3.5 km for medium-capacity lines, with open-ring 22 kV feeds and dual-path supply. ^[9]

The Wenhua record shows how that standard is applied to one medium-capacity line: a 161 kV dual-circuit incoming supply, 22 kV distribution and conversion to 750 V DC at the traction substation before delivery to the conductor rail. ^[10] The exception in this group is the Danhai and Ankeng light rail. NTMC publishes 750 V DC overhead power combined with onboard energy storage, rather than the third rail used by Y and LB. ^[5]

Depots, workshops and service spacing

DORTS identifies Beitou [\[/rail/metro/stations/r22/\]](/rail/metro/stations/r22/) as R's main depot and Xindian [\[/rail/metro/stations/g01/\]](/rail/metro/stations/g01/) as G's. Tucheng [\[/rail/metro/stations/bl03/\]](/rail/metro/stations/bl03/) is the BL main depot with Nangang as a secondary depot; Luzhou [\[/rail/metro/stations/o54/\]](/rail/metro/stations/o54/) is O's main depot, with Xinzhuang [\[/rail/metro/stations/o18/\]](/rail/metro/stations/o18/) secondary and Zhonghe [\[/rail/metro/stations/y12/\]](/rail/metro/stations/y12/) used for storage. ^[11] The planning table separately identifies NeiHu [\[/rail/metro/stations/br19/\]](/rail/metro/stations/br19/) as BR's main depot, Muzha [\[/rail/metro/stations/br02/\]](/rail/metro/stations/br02/) as its secondary depot and South Depot as the first-phase Y main depot. ^[12] Those roles describe depot functions and planned capacity, not a

current train-by-train stabling register. TYMC `[/rail/operators/tymc/]` names Qingpu and Luzhu as A maintenance depots, but its public pages do not map individual Airport MRT sets to either site.^[13]

TRTC publishes a 06:00–24:00 service day for the five high-capacity route families, with separate weekday peak, off-peak, late-evening and holiday headways.^[14] The published weekday peak values are 2–4 minutes on BR, 6 minutes on R with a 3-minute overlap, 4–6 minutes on G with a 3-minute overlap, and 6 minutes on both O and BL with 3-minute overlaps.^[14] NTMC publishes 06:00–24:00 light-rail operation; the Ankeng release gives 6–8-minute morning peaks, 8–10-minute afternoon peaks and roughly 15 minutes at other times.^{[15] [16]}

The station pages already expose first and last trains from the committed TDX `FirstLastTimetable` snapshots.^[17] The repository records the source update and retrieval date in `data/tdx/meta.json`, so these are dated data extracts rather than timeless prose.^[17] The current TDX snapshot contains rows for TRTC, NTMC and TYMC; no TDX rows were returned for the two light-rail operators in the checked extract, so the absence is recorded as data coverage, not as a claim that those operators have no timetable.^[17]

Platform screen doors

The first high-capacity TRTC stations opened without platform doors.^[18] DORTS says new Xinzhuang and Luzhou stations introduced full-height doors and that the older high-capacity stations were retrofitted over a fourteen-year programme, completed in ROC year 107 (2018).^[18] TRTC's 2014 report gives the programme's intermediate state: 58 stations were in the retrofit scope and 13 had been completed by that report, with completion planned by the end of 2018.^[20] DORTS describes Xinyi and Songshan `[/rail/metro/stations/g19/]` as the next full-height-door stage and says the doors were integrated with signalling.^[19]

The Circular Line `[/rail/metro/lines/circular-line/]` has a different equipment module: automatic sliding doors, emergency escape doors, fixed panels and platform-end doors are treated as part of the signalling and interlocking design.^{[21] [4]} The public material fetched for this page does not provide one current, station-by-station TRTC door-type register, and it does not provide a comparable design-capacity-versus-observed-capacity table for every line. Those two fields remain TBC rather than being inferred from neighbouring systems.

What the sources still do not settle

The evidence supports line-level engineering families, published scheduled headways, depot roles and the existing first/last-train data layer.^{[1] [11] [17]} It does not support a current unit-to-depot asset register, a complete station door-type register, or one comparable capacity measure using the same date and unit basis for every line.^{[11] [18] [8]} Those are open data gaps, not missing arithmetic.

Specifications

Taipei Metro service day	06:00–24:00 ^[14]
Wenhu peak headway	2–4 min ^[14]
Tamsui-Xinyi peak headway	6 min, with 3 min overlap ^[14]
Songshan-Xindian peak headway	4–6 min, with 3 min overlap ^[14]
Zhonghe-Xinlu peak headway	6 min, with 3 min overlap ^[14]
Bannan peak headway	6 min, with 3 min overlap ^[14]

References

21 of 21 sources on this page are primary — published by the operator, the builder or government.

- 1** Practical signalling systems for Taipei Metro [<https://ebook.dorts.gov.taipei/ebook/no22/files/basic-html/page73.html>]
捷運號誌系統實務

PRIMARY Taipei City Government Department of Rapid Transit Systems (臺北市政府捷運工程局)
accessed 2026-08-24

Identifies the R, G, O and BL high-capacity signalling family as Alstom (formerly GRS), using conventional fixed blocks and audio-frequency track circuits.
- 2** Practical signalling systems for Taipei Metro — Wenhu Line [<https://ebook.dorts.gov.taipei/ebook/no22/files/basic-html/page225.html>]
捷運號誌系統實務

PRIMARY Taipei City Government Department of Rapid Transit Systems (臺北市政府捷運工程局)
accessed 2026-08-24

Describes the Wenhu Line CITYFLO 650 moving-block coordinate system, roughly half-second position updates and ATP/ATO/ATS.
- 3** Practical signalling systems for Taipei Metro — Circular Line [<https://ebook.dorts.gov.taipei/ebook/no22/files/basic-html/page292.html>]
捷運號誌系統實務

PRIMARY Taipei City Government Department of Rapid Transit Systems (臺北市政府捷運工程局)
accessed 2026-08-24

Identifies Hitachi Rail Transportation Signalling and a driverless CBTC system for the Circular Line, and says the same signalling approach is used on the Sanying Line.

- 4 Practical signalling systems for Taipei Metro — Circular Line equipment [https://ebook.dorts.gov.taipei/ebook/no22/files/basic-html/page303.html]
捷運號誌系統實務
PRIMARY Taipei City Government Department of Rapid Transit Systems (臺北市政府捷運工程局)
accessed 2026-08-24
The detailed Circular Line signalling page identifies interlocking, track circuits and the Faiveley platform-door equipment.
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- 5 Train introduction [https://www.ntmetro.com.tw/basic/?node=10012]
列車介紹
PRIMARY New Taipei Metro Corporation (新北大眾捷運股份有限公司) accessed 2026-08-24
Publishes the Circular, Sanying, Danhai and Ankeng train families, including CBTC, third-rail and overhead/energy-storage power, train formation and operating mode.
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- 6 Planning and design of the Airport MRT train-control system and related interfaces [https://report.ndc.gov.tw/ReportFront/PageSystem/reportFileDownload/C09603079/001]
捷運列車控制系統及相關界面之規劃設計
PRIMARY Ministry of Transportation and Communications (交通部) accessed 2026-08-24
The government report specifies CBTC with ATP, ATS and ATO for the Airport MRT project.
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- 7 Airport MRT electric multiple unit [https://www.trsc.com.tw/train/tran-58acf47251060]
桃園機場捷運電聯車
PRIMARY Taiwan Rolling Stock Co., Ltd. (台灣車輛股份有限公司) accessed 2026-08-24
The manufacturer page gives the Airport MRT train's 750 V DC third-rail supply and CBTC/ATP safety system.
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- 8 Practical power-supply systems for Taipei Metro [https://ebook.dorts.gov.taipei/ebook/no24/files/basic-html/page364.html]
捷運供電系統實務
PRIMARY Taipei City Government Department of Rapid Transit Systems (臺北市政府捷運工程局)
accessed 2026-08-24
Establishes the 750 V DC third-rail standard and the 22 kV transformer/rectifier arrangement, including the distinction between 24-pulse and 12-pulse systems.
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- 9 Practical power-supply systems for Taipei Metro — substations [https://ebook.dorts.gov.taipei/ebook/no24/files/basic-html/page81.html]
捷運供電系統實務
PRIMARY Taipei City Government Department of Rapid Transit Systems (臺北市政府捷運工程局)
accessed 2026-08-24
Gives design spacing for high- and medium-capacity traction substations and the open-ring 22 kV distribution arrangement.
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- 10 Electromechanical planning and design — Wenhua power supply [https://ebook.dorts.gov.taipei/JRTST/ebook/no48/files/basic-html/page139.html]
臺北捷運機電系統精進軌跡尋蹤—機電規劃設計之蛻變、創新與成長
PRIMARY Taipei City Government Department of Rapid Transit Systems (臺北市政府捷運工程局)
accessed 2026-08-24

Details the Wenhua Line's 161 kV dual-circuit incoming supply, 22 kV distribution and conversion to 750 V DC conductor-rail power.

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- 11 Practical metro depot equipment** [<https://ebook.dorts.gov.taipei/ebook/no27/files/basic-html/page27.html>]
捷運機廠設備實務
PRIMARY Taipei City Government Department of Rapid Transit Systems (臺北市政府捷運工程局)
accessed 2026-08-24
Identifies the Beitou, Xindian, Tucheng, Nangang, Luzhou, Xinzhuang and Zhonghe main, secondary and storage functions.
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- 12 Metro network planning practice — depot planning** [<https://ebook.dorts.gov.taipei/ebook/no1/files/basic-html/page202.html>]
捷運路網規劃實務
PRIMARY Taipei City Government Department of Rapid Transit Systems (臺北市政府捷運工程局)
accessed 2026-08-24
The depot table records planning capacities and Wenhua/Circular depot roles; it is not a current unit-to-depot register.
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- 13 Route description** [https://www.tymetro.com.tw/tymetro-new/tw/_pages/travel-guide/road03.php]
路線說明
PRIMARY Taoyuan Metro Corporation (桃園大眾捷運股份有限公司) accessed 2026-08-24
Identifies Qingpu and Luzhu as the Airport MRT maintenance depots.
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- 14 Route and headway information** [<https://www.metro.taipei/cp.aspx?n=ead981369a065968&s=C58C8C2C6419810F>]
路線及班距
PRIMARY Taipei Rapid Transit Corporation (臺北大眾捷運股份有限公司) accessed 2026-08-24
The route pages publish 06:00–24:00 operation and time-of-day headways for all five TRTC route families; the page URL is the Wenhua entry, with the companion line URLs recorded in the research file.
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- 15 Frequently asked questions — service interval and hours** [<https://www.ntmetro.com.tw/basic/?node=10005>]
常見問題
PRIMARY New Taipei Metro Corporation (新北大眾捷運股份有限公司) accessed 2026-08-24
Publishes Danhai/Ankeng light-rail hours, peak and off-peak intervals, branch service patterns and the open-platform operating model.
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- 16 Ankeng LRT commuter ridership grows steadily** [<https://www.ntmetro.com.tw/basic/?mode=detail&node=712>]
安坑輕軌通勤運量穩定增長
PRIMARY New Taipei Metro Corporation (新北大眾捷運股份有限公司) accessed 2026-08-24
Gives Ankeng peak and off-peak service intervals and 06:00–24:00 service hours.
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- 17 **TDX API service documentation** [<https://tdx.transportdata.tw/api-service/swagger/basic/268fc230-2e04-471b-a728-a726167c1cfc>]
TDX API 服務
PRIMARY Ministry of Transportation and Communications (交通部) accessed 2026-08-24
The repository's FirstLastTimetable snapshots are fetched from TDX; data/tdx/meta.json records each operator's source update and retrieval date.
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- 18 **Practical station planning and design — platform doors** [<https://ebook.dorts.gov.taipei/ebook/no2/files/basic-html/page127.html>]
捷運車站規劃與設計實務
PRIMARY Taipei City Government Department of Rapid Transit Systems (臺北市政府捷運工程局) accessed 2026-08-24
Describes the original doorless high-capacity stations, full-height doors on new Xinzhuang/Luzhou stations and the 2006–2018 retrofit programme.
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- 19 **Electromechanical innovations on the Xinyi Line — platform doors** [<https://ebook.dorts.gov.taipei/JRTST/ebook/no51/files/basic-html/page159.html>]
臺北捷運信義線的機電系統創新
PRIMARY Taipei City Government Department of Rapid Transit Systems (臺北市政府捷運工程局) accessed 2026-08-24
Records the full-height platform-door sequence beginning with Xinzhuang/Luzhou and continuing with Xinyi/Songshan, integrated with signalling.
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- 20 **2014 corporate social responsibility report** [<https://web.metro.taipei/ebook/2014ebook/files/basic-html/page47.html>]
2014臺北捷運企業社會責任報告書
PRIMARY Taipei Rapid Transit Corporation (臺北大眾捷運股份有限公司) accessed 2026-08-24
Gives the retrofit baseline of 58 stations, 13 completed by 2014 and the planned end-2018 completion.
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- 21 **Practical signalling systems for Taipei Metro — platform-door modules** [<https://ebook.dorts.gov.taipei/ebook/no22/files/basic-html/page180.html>]
捷運號誌系統實務
PRIMARY Taipei City Government Department of Rapid Transit Systems (臺北市政府捷運工程局) accessed 2026-08-24
Describes the Circular Line's automatic sliding doors, emergency escape doors, fixed panels and platform-end doors.
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Station names, codes and sequence from TDX Transport Data eXchange, Ministry of Transportation and Communications, operators TRTC, NTMC, TYMC, and TMRT. Government open data. Retrieved 2026-08-24; source last updated 2016-11-23. The 12 Sanying station records absent from TDX are sourced directly to New Taipei Metro and government publications on their pages. Provenance.